

Justification Review Document for Other Than Full and Open Competition

Program/Equipment: M109A7 Self-Propelled Howitzer (SPH) - and supporting Carrier Ammunition Tracked (CAT) M992A3 Vehicle production

Authority: 10 United States Code (U.S.C.) 3204(a)(1) as implemented by The Federal Acquisition Regulation (FAR) Part 6.302-1(a)(2)(ii)(A) and (B) -- Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

Amount: BUDGET

Prepared by:

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Date Reviewed: 19-Feb-25

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Date Reviewed: 24-Feb-25

Requirements Representative:

Typed Name: Freeman T. Bonnette

Phone: OPSEC

Title: Colonel, FA
Project Manager
Self-Propelled Howitzer Systems

Date Reviewed: 24-Feb-25

Justification Review Document for Other Than Full and Open Competition

Reviews: I have reviewed this justification and find it adequate to support other than full and open competition.

Program Executive Officer

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DEPARTMENT OF THE ARMY
CLASS JUSTIFICATION AND APPROVAL FOR OTHER THAN FULL AND
OPEN COMPETITION

1. Contracting Activity: Army Contracting Command – Detroit Arsenal (ACC-DTA), Detroit Arsenal, MI.

2. Description of Action: This Class Justification and Approval (J&A) covers multiple contract actions for Project Manager Self Propelled Howitzer Systems (PM SPHS). The contract is for an estimated 281 sets of vehicles that fall under PM SPHS portfolio for M109A7 - Self-Propelled Howitzer (SPH) and M992A3 - Carrier Ammunition Tracked (CAT). The contract will also cover Total Package Fielding (TPF) and Anniston Army Depot (ANAD) support (refurbishment of M109A6 and M992A2 parts to support production of M109A7 and M992A3 vehicles). Award is to BAE Systems Land and Armaments, York, Pennsylvania. The existing Unfinitized Contract Action (UCA) will be modified. The Fiscal Year (FY) 2024 portion of the UCA was awarded under the authority of the prior J&A. The anticipated definitization is not later than 30 September 2025. The second action is for a FY25 UCA, with FY26-28 pricing added at the time of FY25 definitization. This action will result in a five-year Requirements contract.

The contract will include Firm-Fixed-Price, Fixed-Price-Incentive, and Cost Reimbursable contract line item numbers for FY 24-28. If pricing cannot be established for all five years at the time of award, modifications for each funding year will be utilized to add pricing for the out years. Army Procurement Appropriation funds will be obligation. The table in Section 3 summarizes the anticipated quantities of vehicles and support in each FY, as well as the respective amount of funding to support. The first award is planned for the 4QFY26. The contract is a follow-on to contract W56HZV-17-C-0001, which covers production years FY17-23, and was also awarded sole source to BAE for production of the M109A7 and M992A3 vehicles.

3. Description of Supplies/Services: In addition to the estimated 281 sets of vehicles, BAE will supply TPF kits, which consist of spare parts to support the vehicles, and ANAD support from BAE of York, Pennsylvania.

BAE leverages ANAD to refurbish and remanufacture M109A6 and M992A2 components, which are inducted into the M109A7 and M992A3 production lines. The total value of this effort is estimated at **BUDGET** and includes the following:

- i. Up to 281 M109A7s - SPH vehicles
- ii. Up to 281 M992A3s - CAT vehicles
- iii. TPF Kits - up to 10 packages
- iv. ANAD Support

The Independent Government Cost Estimate (IGCE), developed by PM SPHS Business Management in October 2024, summarized the five years of the requirement (including FY24 which is already on contract in an unfinitized state), reflecting an estimated total value of **BUDGET**. For FY24, the IGCE provided an estimate for a quantity of 41 sets of

vehicles; however, the UCA was awarded for a quantity of 40 sets. For FY25-FY28, the IGCE estimated cost for 60 sets of vehicles per FY. The cost summary is below:



4. **Authority Cited:** Title 10 United States Code (U.S.C. 3204(a)(1)), as implemented by Federal Acquisition Regulation (FAR) 6.302-1, Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements. Specifically, the use of this authority for other than full and open competition is supported by FAR 6.302-1(a)(2)(ii)(A) and (B), Supplies may be deemed to be available only from the original source in the case of a follow-on contract for the continued development or production of a major system or highly specialized equipment, including major components thereof, when it is likely that award to any other source would result in (A) Substantial duplication of cost to the Government that is not expected to be recovered through competition; and (B) Unacceptable delays in fulfilling the agency's requirements.

5. **Reason for Authority Cited:** History and current market research supports BAE as the only source capable of meeting the Government's requirement.

Background/History: BAE is the designer and developer of the M109A7 and M992A3 vehicles in their current configuration, originally procured under the W56HZV-14-C-0002 Low-Rate Initial Production (LRIP) contract and Full Rate Production (FRP) contract W56HZV-17-C-0001. BAE has knowledge of the new M109A7 and M992A3 subsystems as well as the predecessor subsystems and how they interface with each other and

function at the overall vehicle system level. Currently, there is no other potential manufacturer that has the unique detailed design and engineering understanding of the M109A7 and the M992A3 vehicles to be able to integrate new, redesigned, and off-the-shelf components into a new chassis while maintaining their performance.

On 03 July 2024, ACC-DTA issued a UCA via Requirements contract W912CH-24-D-0016 and subsequent UCA delivery order W912CH-24-F-0170 with BAE. This award initiates the start of FY24 production which supports ongoing preproduction efforts and authorizes the contractor to produce 40 M109A7, 40 M992A3 vehicles, and two TPF kits. This undefinitized action was authorized via a J&A and Determination and Finding (D&F). A class J&A was approved by the Senior Procurement Executive on 21 August 2020, citing FAR 6.302-1(a)(2)(ii)(A), "Only one responsible source and no other supplies or services will satisfy agency requirements" and on 11 June 2024, the ACC DTA Senior Contracting Official approved the issuance of the requested UCA via a D&F document. After review, he determined the proposed UCA to initiate FY24 SPHS production efforts for 40 M109A7, 40 M992A3 vehicles and TPF kits is in the best interest of the Government, no other contract is suitable and is the only practical method of acquisition available in accordance with FAR 16.603-3, pursuant to Defense Federal Acquisition Regulation Supplement (DFARS) 217.7404-1.

The Government has Level 3 (Product level) Technical Data Packages (TDPs). However, the Government does not have the necessary intellectual property data rights to the pertinent technical data and computer software to enable competitive re-procurement, [REDACTED] OPSEC [REDACTED] For these reasons, the Government is unable to compete the current configuration of the vehicle.

Regarding software, the Government does not have certain key components of the computer software package in its possession. [REDACTED]

[REDACTED] OPSEC [REDACTED]

Per Army Directive 2018-26, and pursuant to the authority of DFARS 252.227-7013(b)(4), 252.227-7014(b)(4), 252.227-7015(c), or 227.7202-3(b), the Government has sought to obtain such technical data, computer software, and data rights that would be sufficient for competitive procurement, several times. Most recently, a request to purchase or license the above was sent to the Contract Administrator at BAE on 23 September 2024 and was declined on 14 October 2024.

Duplication of Cost: BAE is the only source capable of providing the vehicle systems and subsequent support without substantial duplication of costs not expected to be recovered through competition. PM SPHS subject matter experts used program history to make this determination. The chart below provides estimates based on similar efforts performed on

the related SPHS Engineering and Manufacturing Development and System Technical Support contracts.

Activity	Dollars (Million(M))	Paragraph Reference
Engineering Development (Hardware and Software)	BUDGET	i.
Prototype Manufacturing	BUDGET	ii.
Systems Engineering/Program Management	BUDGET	iii.
Systems Test and Evaluation (Developmental Test, Operational Test and Live Fire)	BUDGET	iv.
Other (Small Business Innovation Research/Small Business Technology Transfer)	BUDGET	v.
TOTAL	BUDGET	

i. Engineering Development (Hardware and Software) – The Government does not have data rights to the vehicle that are sufficient for competitive procurement and does not have component level software for the M109 Family of Vehicles (FoVs). The M109 FoV is comprised of highly complex vehicles that require a OPSEC with significant safety considerations that must be taken into account when replacing proprietary designs and software. The estimate includes both the cost to reverse engineer components for which the Government has only limited or restricted rights, the cost to integrate the Line Replaceable Units (LRUs) with the vehicle, and the cost to test the components at both an individual level and the vehicle level. Based on historical data, the redesign and integration of the major LRUs that include

OPSEC

This includes both software and hardware development. The LRU hardware development and software integration is estimated based on current Other Transaction Authority and Work Directives with the contractor. The software development is based on having to rewrite proprietary code and on the learning curve to revise and integrate the remaining code. The estimate also includes the development of Software Integration Labs and outfitting them with testing assets for software validation testing.

ii. Prototype Manufacturing – This estimate includes the cost to build seven prototypes, verify the build process, and modify prototypes based on a test-fix-test scenario. The additional prototypes will validate the integration through a bigger sample and mitigate schedule risk during testing, which is high risk to

compete in such an aggressive timeline. The estimate of BUDGET is based on the actual M109A7 and M992A3 prototype OEM expenditure of BUDGET to build five baseline prototypes and maintain and incorporate test fixes on the prototypes. The remaining BUDGET of the total estimate assumes a BUDGET escalation to account for two additional prototype vehicles, which are needed to meet the compressed time frame for testing and other risk factors.

- iii. Systems Engineering/Program Management – The estimate of BUDGET includes the cost for additional Government staff and contractors to support the engineering development and program management for the duration of the re-engineering efforts, including the reverse engineering efforts to re-design, conduct component qualification, and perform vehicle level testing. These activities are estimated to require six years to complete. This estimate is based on the current engineering support for the program.
- iv. Systems Test and Evaluation (contractor and Other Government Agency) – The estimate of BUDGET accounts for BUDGET in tailored developmental testing, BUDGET in operational testing, BUDGET in contractor test and support, and BUDGET to conduct live fire testing.
- v. Other (Small Business Innovation Research/Small Business Technology Transfer) – The estimate of BUDGET .1M assumes BUDGET of funds will be taken to satisfy the Army's requirement that a portion of funds be set aside/withheld and taxed for every program to support innovative small business and federally funded research and development projects.

Unacceptable Delay: The durations noted in the table below are based on actuals starting from the prototype design through the completion of the Initial Operational Test (IOT). At this point in time, without the ability to provide a complete, competitive production level TDP to another vendor, it is estimated it would take approximately 96 months for a second vendor to be ready to produce FRP vehicles. The 96 months is broken down as follows:

Activity	Duration	Paragraph Reference
Engineering Development (Hardware and Software)	14 months	i.
Prototype Manufacturing	17 months	ii.
Developmental and Limited User Testing on Prototypes	31 months	iii.
Vehicle Build and Facilitization	18 months	iv.
Software Qualification on LRIP Vehicle	2 months	v.
PQT*/FUSL**/IOT	14 months	vi.
TOTAL	96 months	

* Production Qualification Testing (PQT)

** Full-Up System-Level (FUSL)

- i. Engineering Development (Hardware and Software): The 14-month timeline required for the prototype design is attributed to the need to redesign and reengineer the components and systems for which the Government does not have technical data. In order to accomplish this, another contractor would need to develop complete design parameters for components, Computer Aided Design (CAD) models, and performance models for the entire vehicle and each LRU interface. These estimates include reverse engineering of vehicle components, CAD models, and development of performance models and TDPs.
- ii. Prototype Manufacturing: The 17-month timeline is required for the new contractor to acquire vendors, build prototype components and systems, deliver material, and build the prototype vehicles. BAE has a supply chain, with already established procurement lead times, that is facilitating current production builds.
- iii. Developmental and Limited User Testing on Vehicle Prototypes: Any new contractor would be required to build and test prototype vehicles. The vehicle prototypes are required to be tested for reliability and durability to ensure that they meet the vehicle performance specifications. In addition, software qualification testing will be performed on the prototype vehicles. This will take 31 months to complete.
- iv. Vehicle Build and Facilitization: Based on current production build rate and timeline, a new contractor will be required to establish their manufacturing facilities including tooling, fixtures, and inspection tools to build the production vehicles. The timeline includes supply chain development, material lead time, and production build, and is expected to take 18 months.
- v. Software Qualification on LRIP Vehicle: As a basis of estimate, Product Director (PdD) Paladin leveraged the time it took BAE systems to complete the software qualification and assessed that a new vendor would require approximately the same duration of two months to complete the qualification.
- vi. Production Qualification Testing (PQT)/ Full-Up System-Level (FUSL)/IOT: As a basis of estimate, PdD Paladin leveraged the time it took BAE systems to complete the PQT, FUSL Live-Fire test, and IOT, and assessed that a new vendor would require approximately the same duration of 14 months to complete all test activities.

6. Efforts to Obtain Competition:

a. Effective Competition: Based on the detailed rationale provided in paragraph 5 above, the Government has no plans to obtain competition for the proposed contract action at the prime contractor level. In accordance with FAR 5.201, the action will be synopsized on SAM.gov to identify contracting and subcontracting opportunities and to notify potential offerors that all proposals received after the date of publication will be considered by the Government.

b. Subcontracting Competition: This follow-on contract will include the following clauses 52.244-5, Competition in Subcontracting, 52.219-16, Liquidated Damages – Subcontracting Plan, and DFARS 252.219-7003, Small Business Subcontracting Plan (DoD Contracts). In addition, BAE will be required to submit a Small Business Subcontracting Plan as part of its proposals, in accordance with FAR Clause 52.219-9. This plan, along with BAE's bi-annual report submissions, will be used by the Government to monitor the contractor's actual subcontracting activity.

7. Actions to Increase Competition: Since the Government does not have the technical data and/or sufficient data rights to key components, such as the **OPSEC** [REDACTED], the Government is hindered in its ability to facilitate a competitive procurement. The Government will continue to conduct market research and make attempts to obtain specifically negotiated licensed rights in an effort to enhance competition.

8. Market Research: In accordance with FAR 10.002(b)(2), PM SPHS issued a market survey in the form of a Sources Sought Notice to determine the level of interest in this acquisition. The Sources Sought was posted for 30 days to the SAM.gov website, from 16 August 2024 to 16 September 2024. The Sources Sought requested information from vendors regarding their production history, production rate, quality, configuration management, cost, schedule support, supply chain, and life cycle support. No sources responded to express an interest.

9. Interested Sources: To date, no sources have written to express an interest.

10. Other Facts:

a. Procurement History

i. M109A7 and M992A3 Production

(1) Contract number and award date: W912CH24D0016; Undefined Contract Action 03 July 2024

(2) Competition status: Sole Source

(3) Authority previously cited: FAR 6.302-1, Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

(4) Actions to increase competition: Restrictions on the Government's data rights to key components and limitations on delivery of manufacturing data will hinder the Government's ability to compete M109A7 and M992A3 production. The contractor shall make the maximum use of competition between qualified sources. PM SPHS is working to qualify additional sources for key components to the greatest extent practicable, keeping quality and fiscal considerations in mind.

(5) Prior Full and Open Competition: Not applicable.

(6) Unusual Patterns: Not applicable.

(7) Justification for Sole Source Procurement: Award to any other source would result in "unacceptable delays attributable to learning and a substantial duplication of cost to the Government that is not expected to be recovered through competition." "Without the ability to provide a complete, competitive production level TDP to another vendor, it would take approximately 96 months for a second vendor to be ready to produce full rate production vehicles."

ii. M109A7 and M992A3 Production

(1) Contract number and award date: W56HZV-17-C-0001; Unfinalized Contract Action 31 March 2017, Finalized 30 December 2017

(2) Competition status: Sole Source

(3) Authority previously cited: FAR 6.302-1, Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

(4) Actions to increase competition: Restrictions on the Government's data rights to key components and limitations on delivery of manufacturing data will hinder the Government's ability to compete M109A7 and M992A3 production. The contractor shall make the maximum use of competition between qualified sources. PM SPHS is working with the contractor to qualify additional sources for key components to the greatest extent practicable, keeping quality and fiscal considerations in mind.

(5) Prior Full and Open Competition: Not applicable.

(6) Unusual Patterns: FRP decision delayed/postponed several times. FRP granted 24 January 2020.

(7) Justification for Sole Source Procurement: Award to any other source would result in "unacceptable delays attributable to learning and a substantial duplication of cost to the Government that is not expected to be

recovered through competition.” “Without the ability to provide a complete, competitive production level TDP to another vendor, it would take approximately 96 months for a second vendor to be ready to produce full rate production vehicles.” PM SPHS has worked with BAE to qualify additional sources for key components since approval of this J&A. Examples of key components include: the rammer chain links, which were resourced from PROPRIETARY, and the final drive sprocket production which is now split between three machine shops (PROPRIETARY), noting that an additional foundry was added for sprocket production, with PROPRIETARY now casting parts.

iii. LRIP M109A7 and M992A3

(1) Contract number and award date: W56HZV-14-C-0002; 30 October 2013

(2) Competition status: Sole Source to BAE

(3) Authority Previously Cited: FAR 6.302-1(a)(2)(ii), Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

(4) Actions to Increase Competition: The Government intends to require delivery of a Production level TDP as part of the PIM LRIP contract. Restrictions on the Government's data rights to key components and limitations on delivery of manufacturing data (work instructions) will hinder the Government's ability to compete future PIM production.

(5) Prior Full and Open Competition: Not applicable.

(6) Unusual Patterns: None.

(7) Justification for Sole Source Procurement: Award to any other source would result in unacceptable delays attributable to learning and a duplication of cost to the Government that is not likely to be recovered through competition. Since approval of this J&A, the Government acquired a production level TDP; however, it is restricted in its ability to compete the production of the vehicle systems using it due to the number of key contractor components for which it only has limited rights, including the chassis and hull welded structures. The production level TDP also lacks technical data in sufficient detail for some key components to compete the current as-built configurations of the vehicles, including the chassis and hull welded structures.

11. Technical Certification: I certify that the supporting data under my cognizance, which are included in the justification are accurate and complete to the best of my knowledge and belief.

Typed Name: Freeman T. Bonnette

Date: 24-Feb-2025

Title: Colonel, FA
Project Manager
Self-Propelled Howitzer Systems

Signature: BONNETTE.FREE
MAN.TERRILL
Digitally signed by
BONNETTE.FREEMAN.TERRILL
OPSEC
Date: 2025.02.24 17:31:30 -0500

12. Requirements Certification: I certify that the supporting data under my cognizance, which are included in the justification are accurate and complete to the best of my knowledge and belief.

Typed Name: Freeman T. Bonnette

Date: 24-Feb-2025

Title: Colonel, FA
Project Manager

Signature: BONNETTE.FREE
MAN.TERRILL
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BONNETTE.FREEMAN.TERRILL
OPSEC
Date: 2025.02.24 17:31:43 -0500

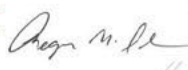
Self-Propelled Howitzer Systems

13. Fair and Reasonable Cost Determination: The contractor's proposal will be reviewed by Government requirements personnel who will address the appropriateness of the proposed vehicle production cost and support ensuring that it is consistent with the IGCE and historical pricing. Cost/Price Analysis and audit recommendations will be used to discern whether the proposed cost/prices are current and justifiable. ACC-DTA has a dedicated Cost and Pricing Division with trained Analysts who will assist the Contracting Officer by performing the necessary analyses to determine fair and reasonableness. Engineers familiar with the program and its systems will be requested to review the proposal for technical validity to ensure that it meets the requirements and is fair and reasonable. Additionally, due to the dollar value of the actions, the Defense Contract Audit Agency and the Defense Contract Management Agency will provide expertise and audit support. The recommendations, as contained in the various analyses, audits, technical reports and historical data will be used by the internal Cost/Price Analyst to develop a negotiating position. This detailed Government negotiation position will be approved under a Business Clearance review at the Defense Pricing and Contracting level. Certified Cost or Pricing Data will be required and form the basis for this analysis. The final negotiated agreement will give full consideration to all contractual requirements, including the delivery schedule, to support a determination of fair and reasonable price for this action. The results of the negotiations and detail of price fairness and reasonableness will be documented in a Post Negotiation Memorandum that will be reviewed and concurred on by the Contracting Officer and at the senior pricing division level.

Typed Name: Regina Ihns

Date: 25-Feb-2025

Title: Contracting Officer

Signature: 
Digitally signed by
IHNS.REGINA.NICHOLE
OPSEC
Date: 2025.02.25 10:48:16 -0500

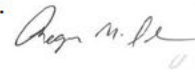
14. Contracting Officer Certification: I certify that this justification is accurate and complete to the best of my knowledge and belief.

Typed Name: Regina Ihns

Date: 25-Feb-2025

Title: Contracting Officer

Signature:

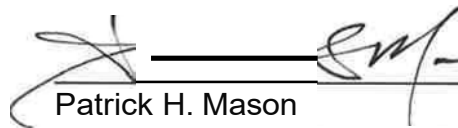


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IHNS, REGINA, NICHOLE
Date: 2025.02.25 10:40:57 -0500

APPROVAL

Based on the foregoing class justification, I hereby approve the procurement of an estimated 281 sets of M109A7 Self Propelled Howitzer (SPH) and M992A3 Carrier, Ammunition, Tracked (CAT) vehicles, Total Package Fielding material, and Anniston Army Depot (ANAD). The total estimated cost is **BUDGET**. Award is to BAE Systems on an other than full and open competition basis pursuant to the authority of Title 10 United States Code 3204(a)(1), as implemented by Federal Acquisition Regulation 6.302-1(a)(2)(ii)(A) and (B), "Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements." This approval is subject to the availability of funds and provided that the supplies and services herein described have otherwise been authorized for this acquisition.

Date: 15 April 2025

A handwritten signature in black ink, appearing to read 'Patrick H. Mason', is written over a horizontal line.

Patrick H. Mason
Senior Official Performing the Duties of the
Assistant Secretary of the Army
(Acquisition, Logistics and Technology)